

MARY SEMERDJIAN

Los Angeles, CA | Email | Portfolio | LinkedIn

EDUCATION

California State University Los Angeles, Engineering, Computer Science and Technology

Los Angeles, CA

Masters of Science in Computer Science

Graduated: May 2024

- **GPA: 3.90/4.00**
Bachelors of Science in Computer Science
- **GPA: 3.24/4.00**
 - Dean's List

Graduated: May 2022

Los Angeles City College, Science, Technology, Engineering, and Math

Los Angeles, CA

Associate of Arts in Computer Science, Computer Information Systems, Liberal Arts - Natural Sciences and Mathematics, Liberal Arts - Arts and Humanities, Associate of Science in Physics for Transfer

Graduated: June 2019

- **GPA: 3.41/4.00**
 - Dean's List

PROFESSIONAL EXPERIENCE

California State University Los Angeles

Los Angeles, CA

Network Security Graduate Assistant

Aug 2023 – Dec 2023

- Supported a graduate-level course, Computer and Network Security course with 60+ graduate students, providing assistance with course material and focusing on networking topics - network security tool.
- Collaborating with the supervisor with additional support and contributing to academic schooling through laboratory work, special projects, and various educational activities; providing students with positive feedback, grading their assignments, and improving the student's understanding of network security and complex cybersecurity concepts.

Master Thesis

Aug 2023 – May 2024

- Authored a Master's Thesis under the supervision of Dr. Ye, focusing on research related to Distributed Denial of Service (DDoS) attacks during my final year as a graduate student.
- Conducted an analysis of detection techniques and mitigation strategies for Distributed Denial of Service (DDoS) attacks by leveraging machine learning, statistical algorithms, probability theory, and neural networks to analyze and employ GPUs for testing effectively and training the datasets.

Capstone Senior Design Program (Biomedical)

Aug 2021 – May 2022

- Learned professional skills in 3D modeling using 3D Slicer and 3D visualization software and experimented with Nvidia AIAA, an API that enables the creation of 3D model objects from trained data, improving automation.
- Focused on creating a custom AI model and training it with our data – MRI images of pelvic organs—utilizing the Clara Train SDK for model training.

TECHNICAL SKILLS | PROJECTS

C++, C, Java, JavaScript, Python, HTML/CSS, PHP, SQL, Haskell, Kotlin, BASIC, Swift, Cisco, Nmap, Metasploit, Snort, Wireshark, Nessus, Jupyter Notebook, 3D Slicer, Scikit-learn, Neural Networks, Keras, MySQL, MariaDB, MongoDB, Bootstrap

Honeypot Server

- Set up a Honeypot server to monitor malicious activity in Kali Linux, and Ubuntu to gain insight into and learn - analyze about the potential attacks and network vulnerabilities.

Histopathologic Cancer Detection

- Tested and analyze different Machine Learning algorithms to create an accurate model that can classify if a given image includes metastatic cancer cells or non- metastatic cancer cells.